

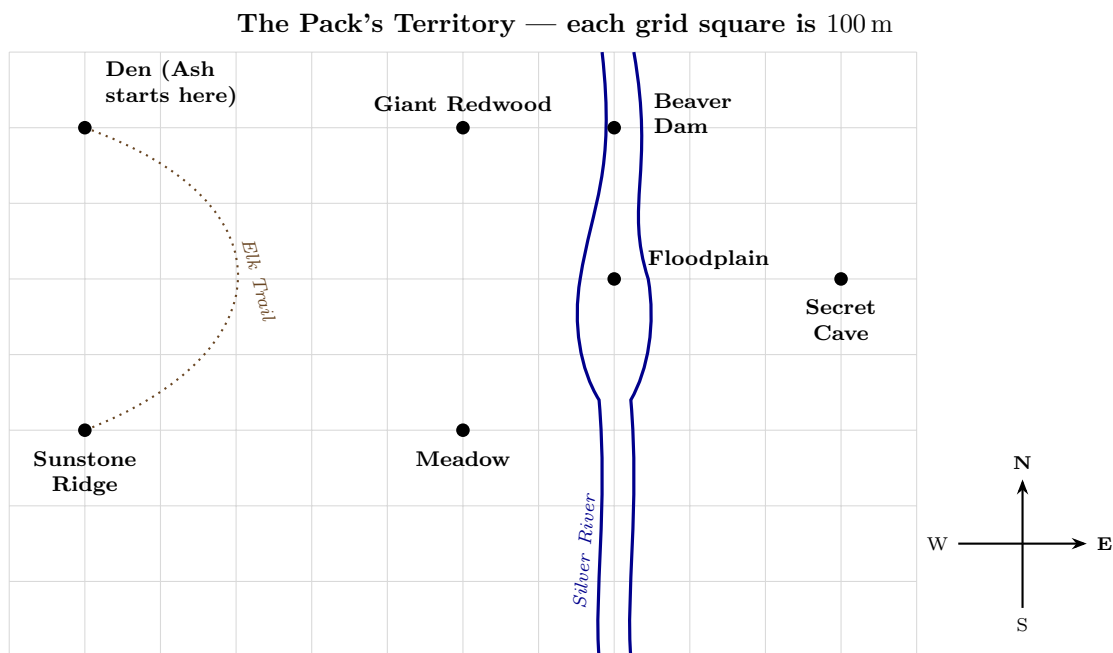
Ash's Journey — Kinematics Review

Grade 6 Physics

Instructions: Ash is a young wolf crossing the pack's territory. The map is divided into 100m squares, and the compass rose shows directions. Use the map to follow Ash's movements. Answer every question; write in complete sentences unless told otherwise, and show your work using the four problem solving steps.

Problem-Solving Steps

1. What am I given? What am I finding?
2. What equation connects them?
3. Plug in and solve.
4. Check the units and direction (if applicable).



Chapter 1 — Waking up at the Den

The pack has already moved on when Ash wakes at the Den, her starting point. The territory lies before her as a grid of 100m squares, and a compass rose shows every direction she will need.

1. Ash's journey begins at the Den, marked on the map. Each grid square is 100 m on a side.
 - (a) What is a **reference point**, and why is the Den a good reference point for Ash's journey?
 - (b) Use the compass and the grid to describe the **position** of the Giant Redwood relative to the Den.

Chapter 2 — Finding Her Way to Beaver Dam

Ash sets off across the territory to reach the Beaver Dam, where the pack has gathered before the crossing. She follows the winding Elk Trail, then works her way along the grid lines.

2. Ash follows the winding Elk Trail from the Den to Sunstone Ridge. The trail is 700 m long, and the trip takes her 10 minutes.

- (a) What **distance** does she travel?

- (b) What is her **displacement** from the Den to Sunstone Ridge? Include the direction.

- (c) Explain why the distance she travels is greater than the magnitude of her displacement.

- (d) Calculate her **average speed** (in $\frac{\text{m}}{\text{s}}$) for this part of the trip. Use $s = \frac{d}{t}$.

- (e) Calculate her **average velocity** (in $\frac{\text{m}}{\text{s}}$) for this part of the trip. Use $\bar{v} = \frac{\Delta x}{t}$.

3. From Sunstone Ridge, Ash walks east to the Meadow in 6 minutes, then north to the Giant Redwood in 8 minutes. She rests at the Redwood for 10 minutes, then walks east to the Beaver Dam in 6 minutes.

- (a) What is the **total distance** she travels from the Den to the Beaver Dam?

- (b) What is her **total displacement** from the Den? Include the direction.

- (c) Explain why the total distance is much larger than the magnitude of the total displacement.
- (d) Calculate her **average speed** (in $\frac{\text{m}}{\text{s}}$) for the whole trip from the Den to the Beaver Dam. Use $s_{\text{avg}} = \frac{d_{\text{total}}}{t_{\text{total}}}$.
- (e) Calculate her **average velocity** (in $\frac{\text{m}}{\text{s}}$) for the whole trip from the Den to the Beaver Dam. Include the direction. Use $\bar{v} = \frac{\Delta x_{\text{total}}}{t_{\text{total}}}$.

Chapter 3 — Closer to the Secret Cave

The morning is still young and the pack is somewhere across the water. At the Beaver Dam, Ash slips into the cold Silver River; from the far bank she has one last run to reach the Secret Cave.

4. At the Beaver Dam, Ash enters the Silver River and swims to the Floodplain, 200 m downstream. In the water she swims at 1.2 m/s south, and the river current flows south at 1.3 m/s.
- (a) What is her **resultant velocity** (magnitude and direction)? Use $v = v_1 + v_2$
- (b) How long does the swim from the Beaver Dam to the Floodplain take? Rearrange $s = \frac{d}{t}$ to solve for t .

5. From the Floodplain, Ash runs east to the Secret Cave, 300 m away. She starts at 10 m/s and, tiring, reaches the cave at 5 m/s. Use

$$a = \frac{v - v_0}{t}, \quad \Delta x = \frac{v + v_0}{2} t.$$

- (a) How long did the run from the Floodplain to the Secret Cave take?

- (b) What was her **acceleration**? Write down its magnitude and direction.

Draw a portrait of Ash in the box below.

One magical protection point will be awarded on the Unit 2 Test, but only if the portrait is completed with genuine effort and demonstrates artistic quality. This protection point allows you to miss one point on the test without it counting against your score. This magic will not work if the drawing is done carelessly.

